

PATTERNS AND SEQUENCES.

Finding next number in the sequence

Note



- The learner should show the sequence used to obtain the next number.
- The operation sign used should be shown.

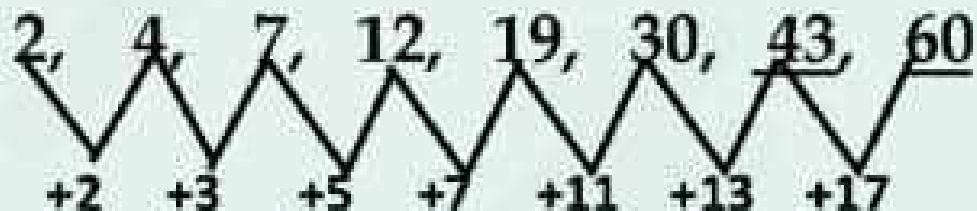
Example

1. Find the next two numbers in the sequence below.

2, 4, 7, 12, 19, 30, _____, _____



Accept the following methods.



Reject

With out showing the sequence.

2, 4, 7, 12, 19, 30, 43, 60

Round off whole numbers.

Note



- Underline the digit in the place to be rounded.
- Look at the digit to the right of the underlined digit.
- If the digit is five or above, give it a share (add 1 to the underlined digit)
- Replace all digits after the underlined digit with zero.

Example

Round off 4263 to the nearest hundreds.



Accept



Reject

TH	H	T	O
4	<u>2</u>	6	3

$$\begin{array}{r} \text{(i) } 4200 \\ + 100 \\ \hline 4300 \end{array}$$

$$\begin{array}{r} \text{(ii) } 4200 \\ + 100 \\ \hline 4300 \end{array}$$

$$\begin{array}{r} \text{(ii) } 4263 \\ + 100 \\ \hline 4300 \end{array}$$

Finding consecutive numbers.

Note



- Emphasise the use of correct sequence depending on the number given.
- Learners should use the given unknown.

Example

- The sum of 3 consecutive even numbers is 42. If the smallest number is $n-3$. Find the numbers.



Accept

1st no	2nd no	3rd no	Sum
$n - 3$	$n - 3 + 2$	$n - 3 + 4$	42

$$n - 3 + n - 1 + n + 1 = 42$$

$$n + n + n - 3 - 1 + 1 = 42$$

$$3n - 3 = 42$$

$$3n = 42 + 3$$

$$3n = 45$$

$$\frac{1}{3} 3n = \frac{1}{3} 45$$

$$n = 15$$

$n - 3$	$n - 1$	$n - 3 + 4$
15 - 3	15 - 1	15 + 1
12	14	16



Reject

1st no	2nd no	3rd no	Sum
n	$n + 2$	$n + 4$	42

$$n + n + 2 + n + 4 = 42$$

$$3n + 6 = 42$$

$$3n = 42 - 6$$

$$\frac{1}{3} 3n = \frac{1}{3} 36$$

$$n = 12$$

1st no	2nd no	3rd no
n	$n + 2$	$n + 4$
12	12 + 2 14	12 + 4 16

Learners did not follow the instructions

Expressing Hindu Arabic numerals to Roman numerals.

Note



- * Roman numerals should be written in capital letters.
- * Roman numerals should not be added.

Examples

1. Express 962 in Roman numerals.



Accept

$$962 = 900 + 60 + 2$$
$$962 = \text{CMLXII}$$



Reject

$$962 = 900 + 60 + 2$$

$$962 = \text{CM} + \text{LX} + \text{II}$$

$$962 = \text{CMLXII}$$

Or

$$962 = 900 + 60 + 2$$

$$= \text{cmixii}$$

(use of small letters)

} Roman
numerals are
not added

SET CONCEPTS

Shading Venn diagrams

Note

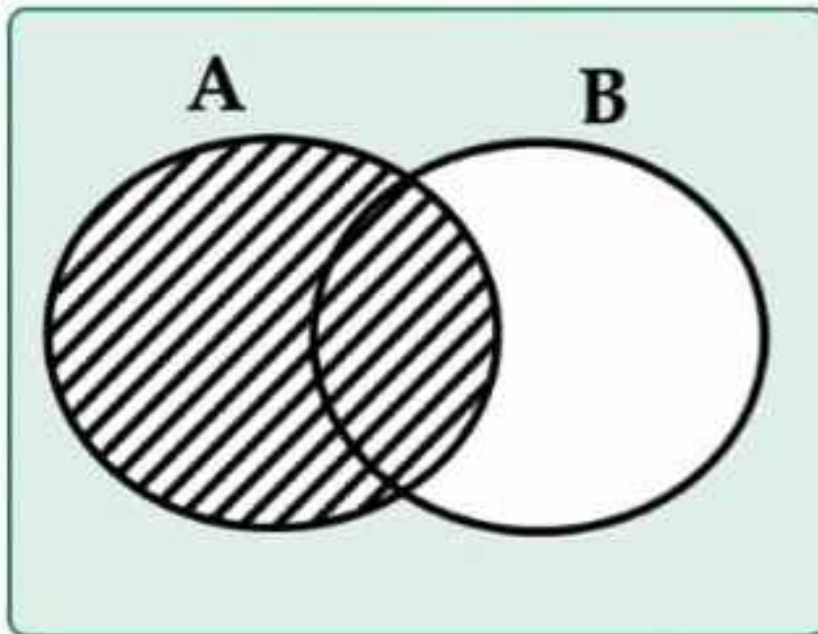


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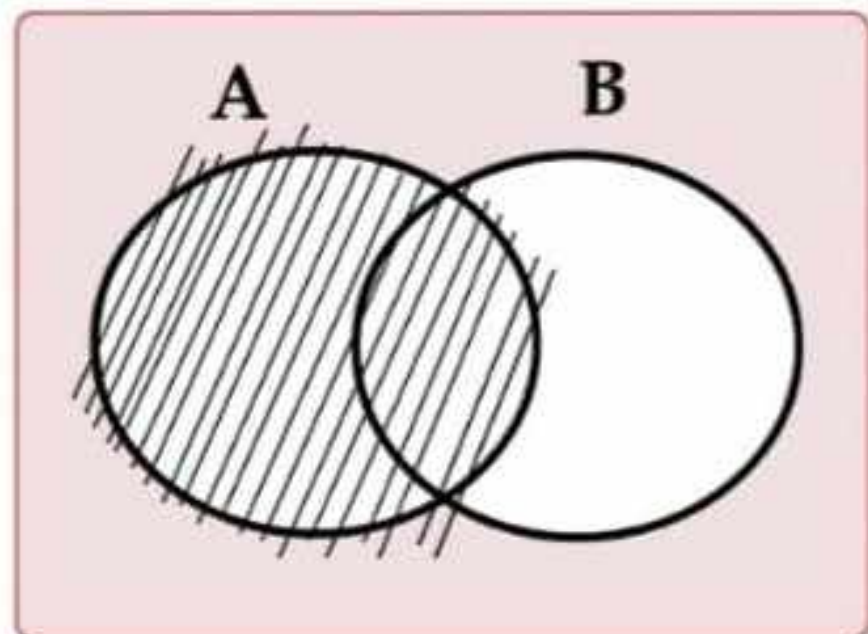
Shade set A.



Accept



Reject



OPERATION ON WHOLE NUMBERS.

Addition of whole numbers.

Note



- The operation sign should be shown.
- The equal signs should be shown.
- Arrange the digits according to place values.

Example.

1. Add $4876 + 273$



Accept

$$\begin{array}{r} 4876 \\ + 273 \\ \hline 5149 \end{array}$$



Reject

$$\begin{array}{r} 4876 \\ 273 \\ \hline 5149 \end{array}$$

Subtraction of whole numbers

Example.

Subtract: $4924 - 3758$

$$\begin{array}{r} 811 \\ 4924 \\ - 3758 \\ \hline 1166 \end{array}$$

SET CONCEPTS

Finding number of elements when given the number of proper sub sets.

Note



- * Encourage correct substitution in the formula.
- * The number of subsets should be prime factorized.

Given that set Y has 15 proper subsets.

Find $n(Y)$



Accept

$$2^n - 1 = 15$$

$$2^n = 15 + 1$$

$$2^n = 16$$

$$2^n = 2^4$$

$$n = 4$$

2	16
2	8
2	4
2	2
	1

$$2^4$$



Reject

Without Prime factorization of sub sets.

$$2^n - 1 = 15$$

$$2^n = 15 + 1$$

$$2^n = 16$$

$$2^n = 2^4$$

$$n = 4$$

Multiplication of whole numbers.

Note

- The operation sign should be shown.
- The equal signs should be shown.

Work out: 426×4



Accept the following methods.

$$\begin{array}{r} 426 \\ \times 4 \\ \hline 1704 \end{array}$$

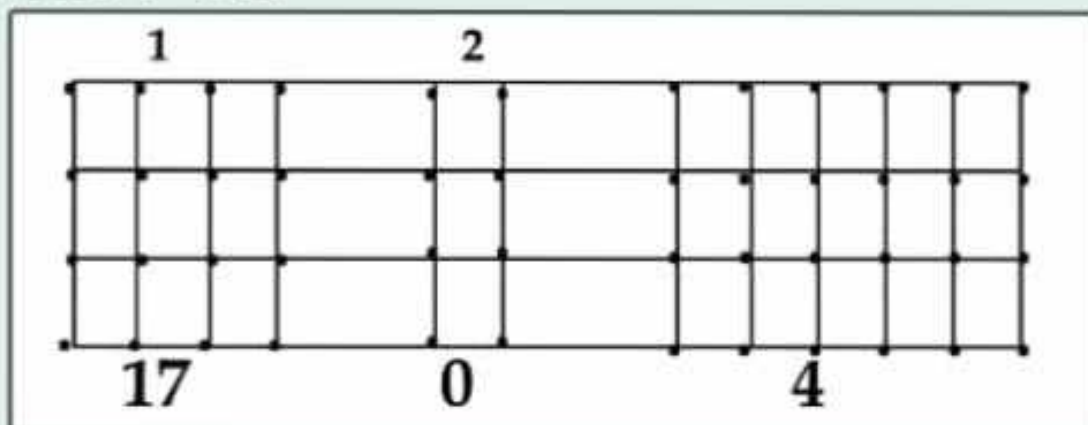
$$\begin{aligned} (6 \times 4) &= 24 \\ (2 \times 4) &= 10 \\ (4 \times 4) + 1 &= 17 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 426 \times 4 &= (400 + 20 + 6) \times 4 \\ &= 1600 + 80 + 24 \\ &= 1704 \end{aligned}$$

(iii) $426 \times 4 = 1704$

x	4	2	6
4	1 6	0 8	2 4
1	7	0	4

(iv) $426 \times 4 = 1704$



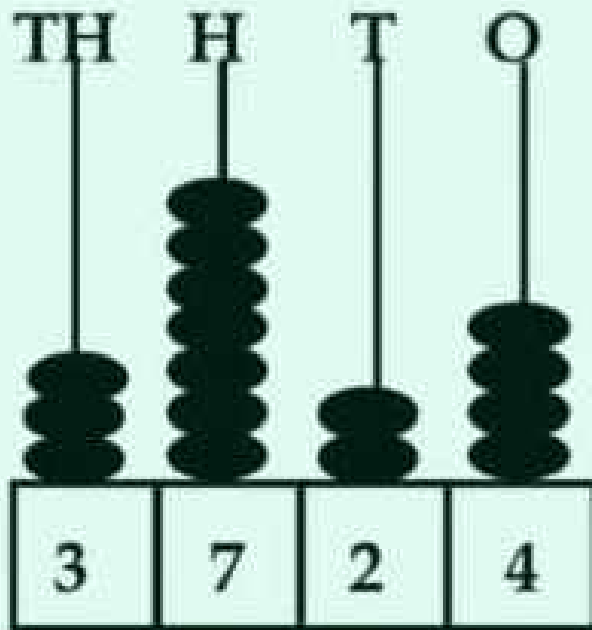
All the above applies to all the operations signs.

The abacus

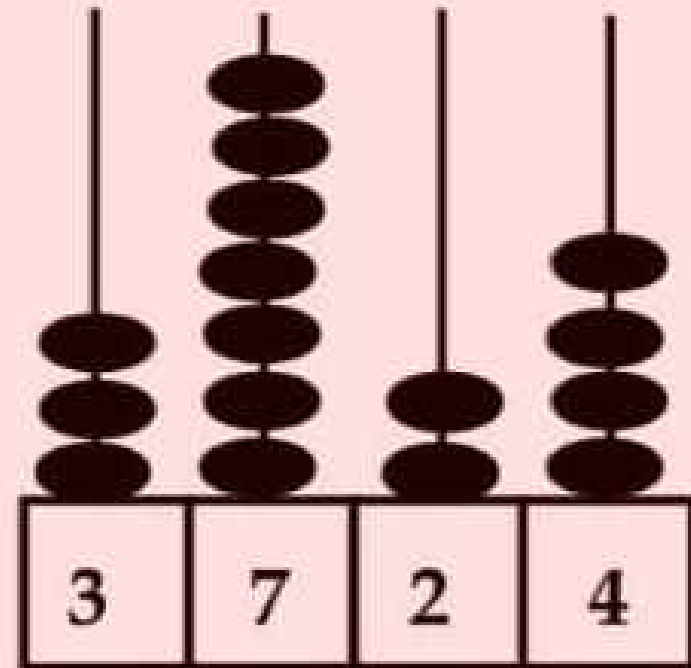
Show 3724 on the abacus.



Accept



Reject



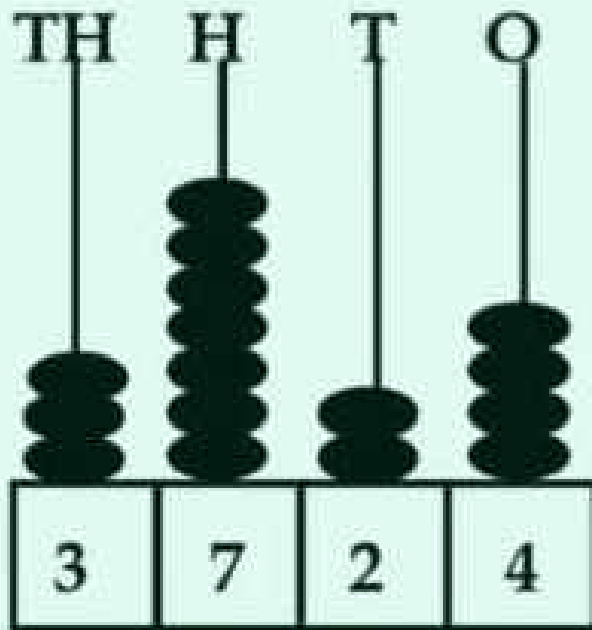
Reject with out place values and when the beads are places wrongly.

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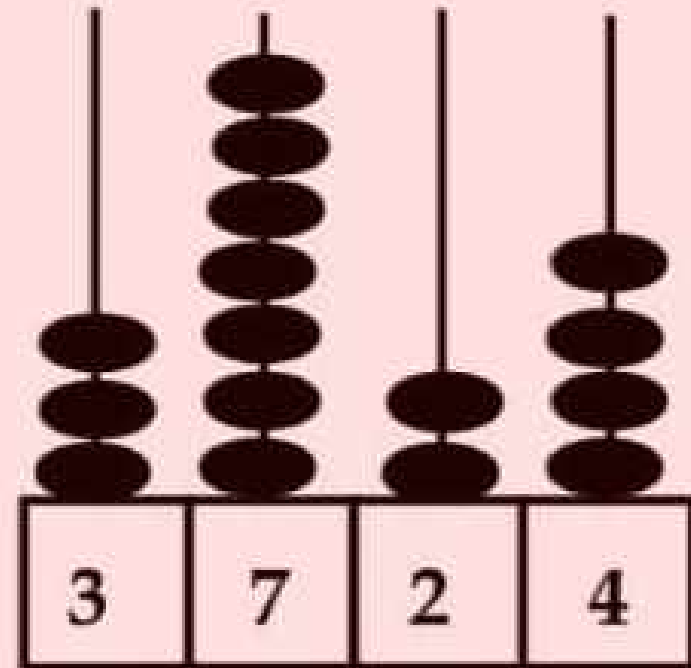
Show 3724 on the abacus.



Accept



Reject



Reject with out place values and when the beads are places wrongly.

FRACTIONS

Converting non-terminating decimals to common fractions

Note



- * Identify the repeating digits in the given decimal number.
- * Equate the decimal number with any variable.
- * Place the digits to the left of the decimal point by multiplying the equation obtained in step 2 by a power of 10 equal to repeated digits.

For example

(a) Convert 0.2626.....to a common fraction.

$$\begin{aligned}\text{(i) Let } P &= 0.2626\ldots\ldots\ldots \\ 100 \times P &= 0.2626\ldots\ldots\ldots \times 100 \\ 100P &= 26.26\ldots\ldots \\ -P &= -0.26 \\ \hline 99P &= 26 \\ 99 & \quad 99 \\ P &= \frac{26}{99}\end{aligned}$$

$$\begin{aligned}\text{(ii) } 0.2626\ldots\ldots \\ \frac{26}{100} - \frac{0}{1} &= \frac{26}{99} \\ \hline \hline\end{aligned}$$

Finding consecutive numbers.

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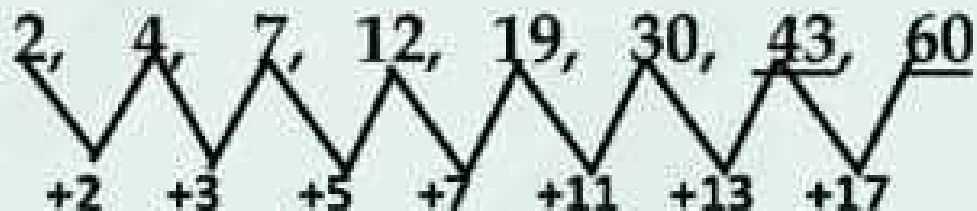
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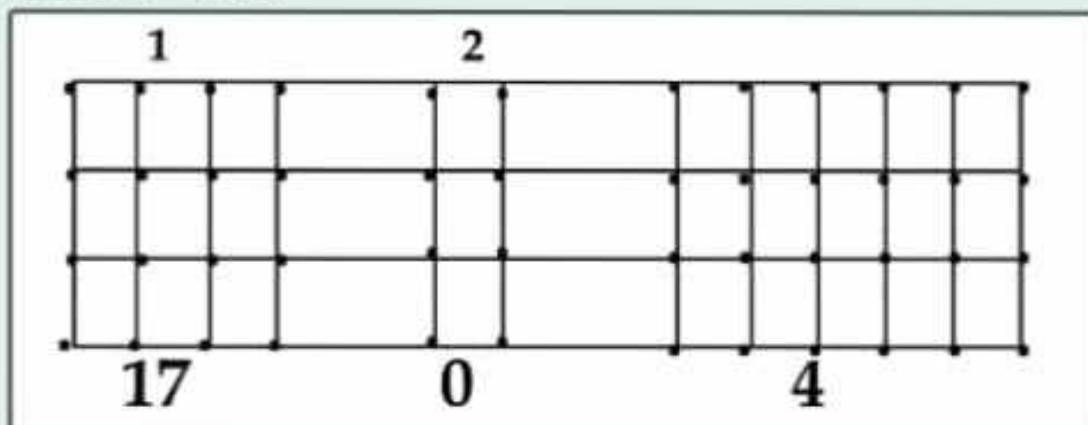
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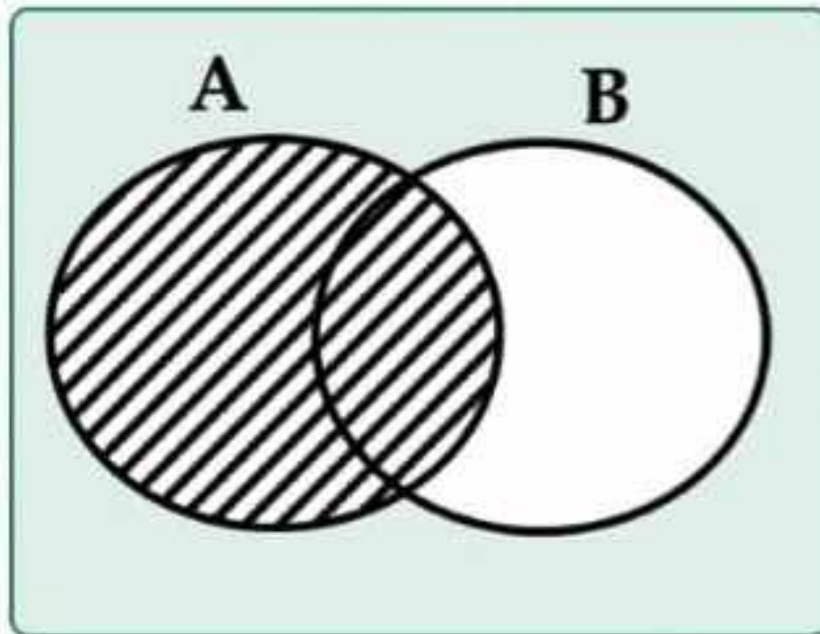


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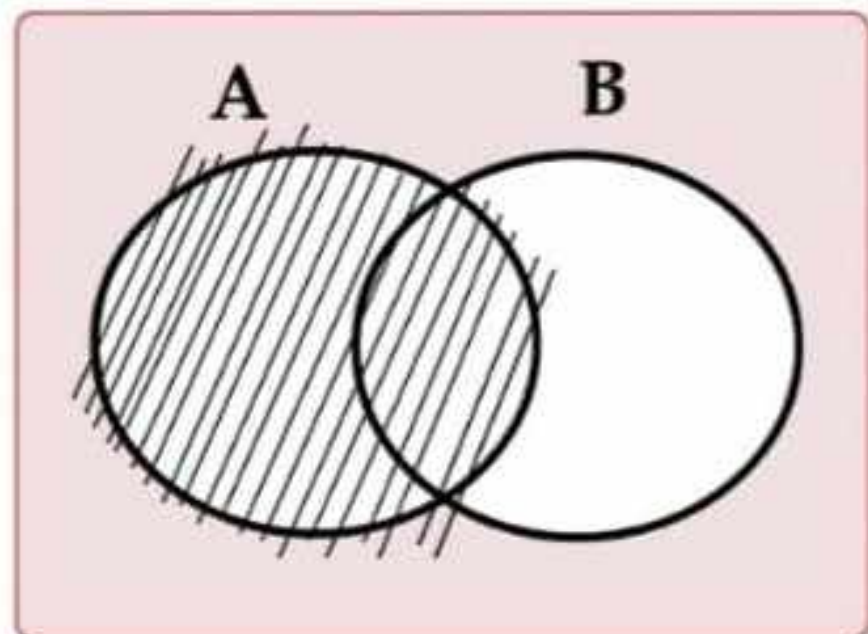
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Completing Venn diagrams

Note

- When completing Venn diagrams, emphasise the use of the given unknowns.
- Learners should not cross while completing the Venn diagram.
- Discourage the final subtraction after finding the value of the unknown.
- Emphasise correct formation of the equation and solving it.

Examples

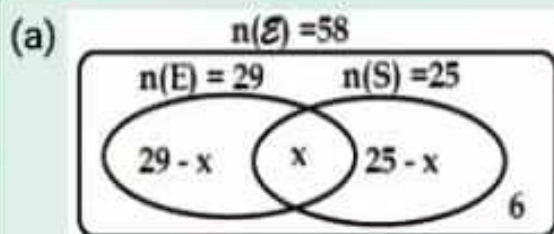
In a class of 58 pupils, 29 pupils like English (E), 25 pupils like Science (S) x pupils like both subjects while 6 pupils like neither of the two subjects.

(a) Represent the above information on the Venn diagram.

(b) Find the value of x .



Accept



(b) Method I

$$29 - x + x + 25 - x + 6 = 58$$

$$29 + 25 + 6 - x = 58$$

$$60 - x = 58$$

$$60 - 58 = x$$

$$2 = x$$

$$x = 2$$

(b) Method II

$$x = (29 + 25 + 6) - 58$$

$$x = 60 - 58$$

$$x = 2$$

(b) Method III

$$29 + 25 - x + 6 = 58$$

$$60 - x = 58$$

$$60 - 58 = x$$

$$2 = x$$

$$x = 2$$

(b) Method IV

$$25 + 29 - x + 6 = 58$$

$$60 - x = 58$$

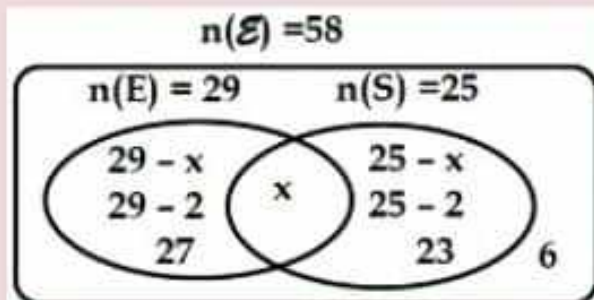
$$60 - 58 = x$$

$$2 = x$$

$$x = 2$$



Discourage



Reject

$$29 - x + x + 25 - x + 6 = 58$$

$$29 + 25 + 6 - x = 58$$

$$60 - x = 58$$

$$-x = 58 - 60$$

$$\begin{array}{c} +x \\ +x \end{array} = \begin{array}{c} +2 \\ +x \end{array}$$

$$x = 2$$